

A Note on Cliques and Independent Sets

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Abstract

For integers $m, n \geq 2$, let $f(m, n)$ be the minimum order of a graph where every vertex is in both a clique of cardinality m and an independent set of cardinality n . We show that $f(m, n) = \lceil (\sqrt{m-1} + \sqrt{n-1})^2 \rceil$.

In this paper we use the terminology of [1]. Specifically, graph G has vertex set $V(G)$, edge set $E(G)$ and order $p = |V(G)|$. Also, $N(v)$ denotes the neighborhood of a vertex v , i.e., the set of all vertices adjacent to v . Further, an m -clique is a complete subgraph of order m .

For positive integers m and n , let $f(m, n)$ be the minimum order of a graph where every vertex is in a clique of cardinality m , and every vertex is in an independent set of cardinality n .

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Theorem 1 For $n, m \geq 2$,

$$f(m, n) = n + m - 2 + \left\lceil 2\sqrt{(m-1)(n-1)} \right\rceil.$$

We start with some lemmas needed in the proof of the lower bound. We say that a graph is *m-clique-minimal* if every vertex is in an *m*-clique, but the removal of any edge leaves some vertex not in an *m*-clique. Let *U* denote the set of vertices in unique cliques (equivalent to degree $m-1$ in such graphs).

Lemma 1 Let *G* be *m-clique-minimal*. Then for every edge *e* of *G* there is a vertex *U* whose unique *m*-clique contains *e*.

Proof. We say that an edge is *critical* to the vertex *v* if the removal of that edge destroys all *m*-cliques containing *v*. Since *G* is *m-clique-minimal* the edge *e* is critical to some vertex. Call it y_1 .

Let F_1 be an *m*-clique containing y_1 . Necessarily F_1 contains *e* and hence *v* and *w*. If F_1 is unique, then we are done: set $u = y_1$. Otherwise there is an edge y_1x_2 such that $x_2 \notin F_1$. The edge y_1x_2 is critical to some vertex, say y_2 . (Of course, perhaps $y_2 = x_2$.) Note that $y_2 \notin \{v, w, y_1\}$. But every *m*-clique that contains y_1x_2 contains y_1 , and so contains *e*. Hence *e* is critical to y_2 . Indeed, every *m*-clique containing y_2 contains y_1 , *v* and *w* as well.

Let F_2 be an *m*-clique containing y_2 . If F_2 is not unique we find an edge y_2x_3 with $x_3 \notin F_2$, and, by similar reasoning to the above, a vertex y_3 such that every *m*-clique containing y_3 contains y_2 , y_1 , *v* and *w* also. If F_i is ever unique we are done: set $u = y_i$. Otherwise we produce an infinite sequence $y_1, y_2, y_3, \dots$ such that every *m*-clique containing y_i contains $y_{i-1}, y_{i-2}, \dots, y_1, v$ and *w*. But this is absurd: y_{m-1} would be in an *m*-clique with *m* other vertices. This establishes the lemma. $\square$

For vertex *v*, let $\beta(v)$ denote the maximum cardinality of an independent set containing *v*.

Lemma 2 Let *G* be *m-clique-minimal*. If *S* is a maximum independent subset of *U*, then $\beta(v) = |S - N(v)| + 1$.

Proof. Certainly $\{v\} \cup (S - N(v))$ is independent and contains *v*. Thus $\beta(v) \geq |S - N(v)| + 1$.

Now, amongst all the independent sets of cardinality $\beta(v)$ that contain *v*, let *T* be the one with $|T - S|$ as small as possible. Suppose $w \in T - S$. By Lemma 1

applied to an edge incident with w , there is a vertex $u \in U$ such that w is in the unique m -clique containing u . (Perhaps $u = w$.) If $u \in S$ then we may replace w by u in T and obtain an independent set since $N(w) \supseteq N(u)$. But this independent set contradicts our choice of T . If $u \notin S$, then by the maximality of S , u has a neighbor u' in S ; and w is in the m -clique containing u' . So we may replace w by u' in T and obtain a contradiction as before. $\square$

Lemma 3 *Let G be m -clique-minimal and n -independent. Then $p \geq n + m - 2 + \left\lceil 2\sqrt{(m-1)(n-1)} \right\rceil$.*

Proof. Let S be a maximum independent subset of U and let $X = V(G) - S$ and $x = |X|$. Consider $B = \sum_{v \in X} \beta(v)$. Lemma 2 implies that $B = x + \sum_{v \in X} |S - N(v)|$. But every vertex w of S is adjacent to exactly $m - 1$ of the vertices in X . Thus $B = x + |S|(|X| - (m - 1)) = (p - x)(x - (m - 1))$. Since G is n -independent, it follows that $B \geq nx$. Thus

$$x + (x - (m - 1))(p - x) \geq nx.$$

Rearranged, the inequality

$$p \geq (n - 1) + (m - 1) + \frac{(m - 1)(n - 1)}{\alpha} + \alpha,$$

where $\alpha = x - (m - 1)$. Since the function $\alpha + z/\alpha$ assumes its minimum value at $\alpha = \sqrt{z}$, it follows that

$$p \geq n + m - 2 + \left\lceil 2\sqrt{(m - 1)(n - 1)} \right\rceil$$

vertices. $\square$

We now show the upper bound by constructing a suitable graph as a member of a more general class. Given positive integers m, x, s where $s(m - 1) \geq x$ and $x \geq m - 1$, we define a graph $G(m, x, s)$. The vertex set is partitioned into a clique W of cardinality x and an independent set U of cardinality s . We add $s(m - 1)$ edges with one end in W and the other end in U distributed evenly amongst W . In particular, let $W = \{w_0, w_1, \dots, w_{x-1}\}$ and $U = \{u_1, u_2, \dots, u_s\}$. Then for $1 \leq i \leq s$ join u_i to the $m - 1$ vertices w_j for $(m - 1)(i - 1) \leq j \leq (m - 1)i - 1$, where subscripts are read modulo x .

We have explicitly placed each vertex of U in an m -clique, and these cliques cover W . Furthermore, each vertex of W has at most $\lceil s(m - 1)/x \rceil$ neighbours in W . Thus

each vertex of W belongs to an independent set of cardinality at least $s - \lceil s(m-1)/x \rceil + 1$, and these sets cover W .

Given m, n , set $\alpha = \sqrt{(m-1)(n-1)}$, $s = n-1 + \lceil \alpha \rceil$ and $x = m-1 + \lceil \alpha \rceil$ where $\lceil \alpha \rceil$ denotes the integer nearest to α . Since $m, n \geq 2$, we have both $x \geq m-1$ and $s(m-1) \geq x$. Since $\lceil \beta \rceil + \lceil \beta \rceil = \lceil 2\beta \rceil$ for any real number β , the graph has order $s + x = (n-1) + (m-1) + \lceil 2\alpha \rceil$.

It remains only to verify that $s - \lceil s(m-1)/x \rceil + 1 \geq n$ or, equivalently, that

$$\lceil \alpha \rceil \geq \left\lceil \frac{(n-1 + \lceil \alpha \rceil)(m-1)}{m-1 + \lceil \alpha \rceil} \right\rceil. \quad (1)$$

Let $y = \lceil \alpha \rceil$. If $\lceil \alpha \rceil = y$, then since $y \in \text{integers}$ it suffices to prove that $y \geq (n-1+y)(m-1)/(m-1+y)$, which simplifies to $y^2 \geq (m-1)(n-1)$ and hence to $y \geq \alpha$, which is true. On the other hand, if $\lceil \alpha \rceil = y+1$, then since $y+1 \in \mathbf{Z}$ it suffices to prove that $y+1 \geq (n+y)(m-1)/(m-1+y)$, which simplifies to $y^2 + y \geq (m-1)(n-1)$. Since $y \geq \alpha - 1/2$, it follows that $y^2 + y \geq \alpha^2 - 1/4 = (m-1)(n-1) - 1/4$. Since $y, m, n \in \text{integers}$, the desired inequality holds.

This completes the proof of Theorem 1.

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